Do Now · retrieval

Lesson 094 · CPU Performance: Clock, Cores and Cache

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a cache hit?

Q2

What is a cycle?

Q3

What is a core?

Q4

A search space of 64 candidates is repeatedly halved to one. How many halvings?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Clock speed counts clock cycles per second; one cycle is not always one complete instruction. Frequency alone cannot predict program performance.

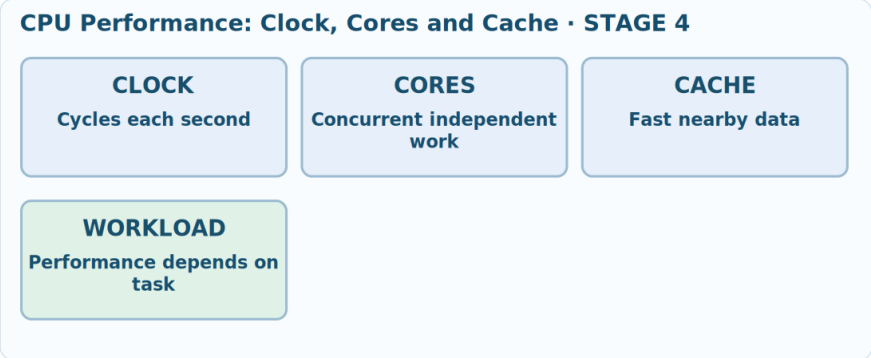
Core idea 2

Explain — Multiple CPU cores improve performance most for work divided into independently executable threads; a strictly sequential task may gain little.

Core idea 3

Evaluate — Cache is small, fast storage close to CPU. A cache hit avoids some slower RAM accesses; cache size, locality and hit rate influence results.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

GHz — Billions of cycles per second.

core — CPU execution core.

cache hit — Data found in CPU cache.

IPC — Instructions retired per cycle (workload dependent).

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

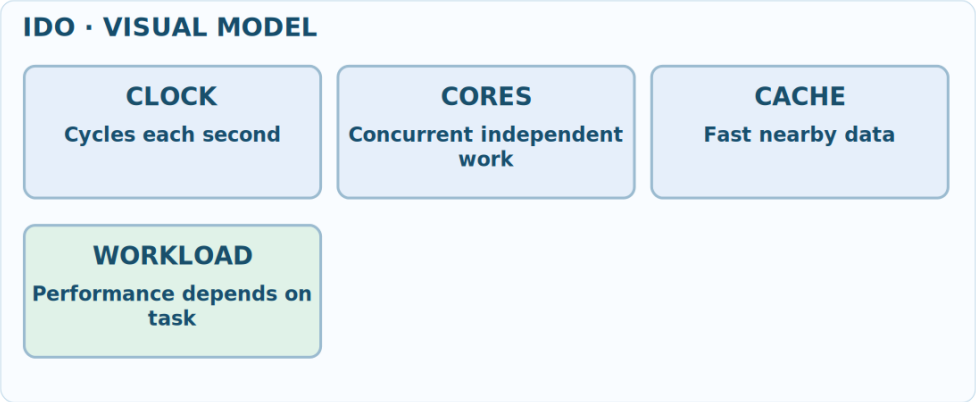
Problem

Compare 2 GHz and 3 GHz clock rates under equal work per clock and no bottlenecks.

Follow the worked method

1. Explain the mechanism — 3 GHz offers 50% more cycles each second than2 GHz; this does not universally imply 50% faster program runtime. — The comparison requires equal microarchitecture and workload assumptions.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the mechanism
It may not be substantially faster from extra cores alone; only one thread is the bottleneck.
Parallel benefits need tasks/threads that can run concurrently.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Explain effects of clock speed, number of cores and cache on performance. [6]

Question 2

CPU X 2GHz and Y 3GHz have identical IPC; compute ratio of cycles/sec and one reason runtime may differ from ratio. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Recommend improvements for school video rendering versus a sequential exam-marking script. Discuss CPU performance factors. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A 4GHz processor always completes any program twice as fast as a 2GHz processor.

Use key vocabulary from memory

1. Explain GHz in your own words.

2. Explain core in your own words.

3. Explain cache hit in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.